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Simultaneous Guillain-Barré Syndrome and Acute Disseminated Encephalomyelitis in the Pediatric Population

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Few cases of simultaneous acute demyelination of the peripheral and central nervous systems are reported. Four patients diagnosed as having Guillain-Barré syndrome and acute disseminated encephalomyelitis during the same hospitalization are described herein. Two patients manifest an atypical form of Guillain-Barré syndrome, with magnetic resonance imaging of the head showing acute disseminated encephalomyelitis. A third patient has acute disseminated encephalomyelitis and develops Guillain-Barré syndrome during his hospitalization. A fourth patient demonstrates transverse myelitis that evolves into Guillain-Barré syndrome, with demyelination seen on brain magnetic resonance imaging. All patients are treated with intravenous immunoglobulins or corticosteroids. Three patients have a favorable outcome; 1 patient has a chronic inflammatory demyelinating polyradiculoneuropathy. Guillain-Barré syndrome and acute disseminated encephalomyelitis can occur simultaneously in the pediatric population. This may be explained by a shared epitope between peripheral and central nervous system myelin. Further research is necessary to better describe this entity and its prognosis.

Keywords: demyelination; Guillain-Barré syndrome; inflammatory demyelinating polyradiculoneuropathy; acute disseminated encephalomyelitis

Guillain-Barré syndrome, or acute demyelinating polyradiculoneuropathy, and acute disseminated encephalomyelitis are 2 common and well-defined diseases of the pediatric population. Both are hypothesized to be related to a viral-induced autoimmune response. The occurrence of these 2 entities, simultaneously or sequentially during a short period, is not well understood and is probably underdiagnosed. We review the hallmarks of 4 pediatric cases of simultaneous or sequential Guillain-Barré syndrome and acute disseminated encephalomyelitis during a short period.

Methods

Between 2003 and 2006, 4 patients presented to our institution with combined demyelination of the peripheral nervous system and the central nervous system. A medical record review of these 4 patients was undertaken to better define this potential entity.

Results

A summary of the clinical presentation, investigations, treatment, and prognosis for each of the 4 patients is given in Table 1. Each case is presented in detail herein.

Case 1

Our first case was a 10-year-old boy, previously healthy, who presented to our institution with an altered level of consciousness, a T5 sensory level, diffuse weakness (lower extremities were more affected), hyperreflexia, bilateral Babinski signs, and urinary and fecal retention. The patient had experienced an upper respiratory tract infection 3 weeks before presentation and viral gastroenteritis 3 days before admission. Magnetic resonance imaging of the spine and brain showed multiple signal abnormalities in the white matter, consistent with acute disseminated encephalomyelitis. Lumbar puncture revealed mild pleocytosis (leucocyte count of 60/μL, with 45% neutrophils, 20% lymphocytes, and 35% monocytes) and a high protein level (0.70 g/L). The patient

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Table 1. Summary of Cases

Variable	Case 1	Case 2	Case 3	Case 4
Age, y	10	5	11	16
Prior infectious illness	Upper respiratory tract infection, gastroenteritis, *Mycoplasma*	Epstein-Barr virus	Upper respiratory tract infection	Gastroenteritis
Primary diagnosis	Acute disseminated encephalomyelitis	Acute disseminated encephalomyelitis	Guillain-Barré syndrome	Transverse myelitis, Guillain-Barré syndrome
Secondary diagnosis	Guillain-Barré syndrome	Guillain-Barré syndrome (mild), chronic inflammatory demyelinating polyradiculoneuropathy	Acute disseminated encephalomyelitis	Acute disseminated encephalomyelitis (?)
Electromyography and nerve conduction study findings	Acute demyelinating polyradiculoneuropathy	Chronic changes, proximal root demyelination	Acute demyelinating and axonal polyradiculoneuropathy	Conduction block, absent F waves
Treatment	Methylprednisone, intravenous immunoglobulins	Methylprednisone	Intravenous immunoglobulins, gabapentin	Intravenous immunoglobulins
Expanded Disability Status Scale score[a]				
Worst	7-9	5-6.5	7-9	5-6.5
At discharge	5-6.5	0-2.5	3-4.5	3-4.5
At last follow-up	0-2.5	0-2.5	0-2.5	0-2.5
Overall prognosis	Very good	Very good	Very good	Good

a. Score of 0 to 2.5 indicates normal neurologic examination results or mild abnormal signs without disability; 3 to 4.5, abnormal signs plus some degree of disability but fully ambulatory without aid; 5 to 6.5, severely disabled with unilateral or bilateral assisted walking; 7-9, unable to walk and restricted to wheelchair or bed; and 10, death due to demyelinating disease.

was treated with methylprednisolone (1 g/d for 5 days), with mild improvement. During his hospitalization, he developed increasing weakness and hyporeflexia. Nerve conduction studies revealed prolonged motor and sensory latencies, slowed motor and sensory nerve conduction velocities, and prolonged or absent F waves. Needle examination revealed a small amount of fibrillation potentials, motor unit action potentials of increased amplitudes and duration, and mildly decreased recruitment. These results were compatible with the diagnosis of Guillain-Barré syndrome. A 3-dose course of intravenous immunoglobulins was administered, which resulted in significant improvement of his weakness. Serologic examination for *Mycoplasma* was positive, but the cerebrospinal fluid polymerase chain reaction was negative. The remaining serologic studies and cultures were negative. He was discharged from the hospital 39 days after his admission, ambulating with help and able to transfer but still requiring intermittent catheterization. Six months after presentation, magnetic resonance imaging of the brain and spine showed complete resolution of the signal abnormalities, and neurologic follow-up showed no residual deficits (except for bilateral Babinski reflexes).

In summary, case 1 presented with typical acute disseminated encephalomyelitis and during his hospitalization developed Guillain-Barré syndrome. He was treated sequentially with intravenous methylprednisone and intravenous immunoglobulins, with significant improvement.

Case 2

Our second patient was a 5-year-old Native American girl. She presented to our institution with diffuse progressive weakness (predominantly proximal), areflexia, and dysesthesias 1 week after a viral illness. Lumbar puncture was performed early in the course and revealed a normal protein level. Serologic findings were compatible with acute Epstein-Barr virus infection. Magnetic resonance imaging of the spine showed enhancement of the ventral and dorsal roots and of the cauda equina, which was compatible with Guillain-Barré syndrome. Nerve conduction studies and electromyography were performed, with normal findings. Magnetic resonance imaging of the brain was performed because of the atypical presentation (proximal more than distal weakness and normal electrophysiologic findings) and showed several small areas of abnormal signal in the white matter, compatible with a diagnosis of acute disseminated encephalomyelitis. She was treated with intravenous methylprednisolone for 5 days, followed by a 2-week prednisone taper. Intravenous immunoglobulins were not given because of preserved ambulation. She

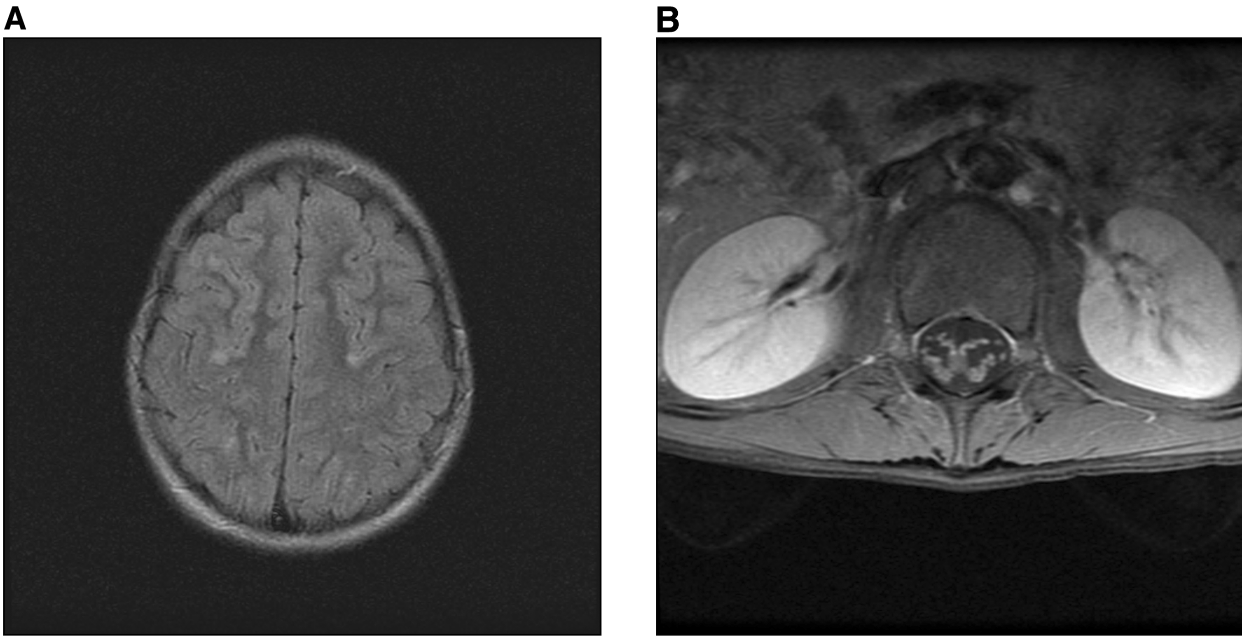

Figure 1. T1-weighted axial magnetic resonance images from case 3. (A) Brain showing several subcortical and white matter round demyelinating lesions. (B), Spine after gadolinium administration showing enhancement of the roots and cauda equina.

improved significantly and retained only mild proximal weakness. During the next year, she presented twice with proximal more than distal limb weakness and hyporeflexia. Results of nerve conduction studies performed on these 2 occasions suggested proximal root demyelination, and needle examination revealed positive sharp waves, fibrillation potentials, and polyphasic motor unit action potentials of larger amplitude and prolonged duration. Together with the clinical presentation, these results suggested chronic inflammatory demyelinating polyradiculoneuropathy. The patient was started on intravenous immunoglobulins and responded well.

In summary, case 2 initially presented with a clinical diagnosis of Guillain-Barré syndrome associated with mild acute disseminated encephalomyelitis, as evidenced by magnetic resonance imaging of the spinal cord and brain. Although she initially improved significantly with intravenous methylprednisolone, she eventually went on to develop chronic inflammatory demyelinating polyradiculoneuropathy that responded well to intravenous immunoglobulins.

Case 3

Our third patient was an 11-year-old boy of white race/ethnicity, previously healthy, who presented with progressive lower limb weakness, right facial paresis (lower motor neuron type), and decreased deep tendon reflexes following an upper respiratory tract infection. During a few hours, he developed quadriparesis, facial diplegia (lower motor neuron type), sensory disturbances, autonomic instability, increased secretions, and dysphagia. He was transferred to the intensive care unit and was intubated for airway protection. He remained intubated for 4 days. Lumbar puncture was performed and revealed a high protein level (2.9 g/L). Nerve conduction studies showed increased motor latencies, decreased motor amplitudes, no sensory potential in the upper extremities, and absent F wave in the lower extremities. Concentric needle examination revealed mildly reduced recruitment, with slightly larger motor unit action potentials. These results confirmed the diagnosis of acute demyelinating and axonal polyradiculoneuropathy. Magnetic resonance imaging of the spinal cord revealed enhancement of the ventral and dorsal roots and of the cauda equina, compatible with a diagnosis of Guillain-Barré syndrome. Significant improvement was noted with intravenous immunoglobulin therapy. Gabapentin was started for neuropathic pain. In view of the findings of mild psychomotor slowing (confirmed by neuropsychologic assessment) and preserved stretch reflexes, despite severe weakness, a decision was made to perform brain magnetic resonance imaging. This revealed several small areas of abnormal signal affecting the subcortical white matter of both hemispheres, consistent with the diagnosis of acute demyelinating encephalomyelitis (Figure). Because of the mild central nervous system symptoms, a decision was made to observe the patient only, and no corticosteroids were given. The patient improved during his hospitalization and was discharged 15 days later with mild facial diplegia, mild difficulty with toe walking, positive Gower sign, and some instability with the Romberg maneuver. Otherwise, he was walking and transferring independently, climbing stairs with help, and hopping on 1 foot bilaterally. Two months later, the patient had recovered almost completely except for mild right facial plegia and minimal weakness of the deltoids. Neuropathic pain resolved, and gabapentin was stopped. Magnetic resonance imaging of the spine and brain showed complete resolution of the radicular enhancement and the brain white matter abnormalities.

In summary, case 3 presented with rapidly progressive Guillain-Barré syndrome with mildly symptomatic acute disseminated encephalomyelitis. He had almost complete recovery with intravenous immunoglobulins.

Case 4

Our fourth patient was a 16-year-old boy of white race/ethnicity, previously healthy, who presented after a prodrome of gastroenteritis with bilateral leg weakness and pain, urinary and fecal retention, and a T8 to T10 sensory level. A diagnosis of transverse myelitis was made on the basis of the clinical picture. Magnetic resonance images of the head and spine were normal. Lumbar puncture was notable only for a high protein level (0.85 g/L). Significant improvement of all deficits occurred during the following days. As he was recovering, he suddenly developed facial diplegia. Within the next 24 hours, he developed sensory deficits over the mandibular branch of the fifth cranial nerve bilaterally, left partial third nerve palsy, and left fourth nerve palsy. Later on, he developed dysesthesias in the tips of his fingers, hyporeflexia, worsening weakness, and mild autonomic instability. Nerve conduction studies revealed conduction block at the knee for the left peroneal nerve and absent left peroneal F wave. Needle examination showed slightly reduced recruitment and mild increased amplitude of the motor unit action potentials with some polyphasic units. Together with the clinical presentation, these results confirmed the diagnosis of Guillain-Barré syndrome. Magnetic resonance imaging of the spine showed diffuse radicular enhancement. A second lumbar puncture showed a dramatic increase in the protein level (5.1 g/L). The patient improved significantly with administration of intravenous immunoglobulins. During his hospitalization, he developed bilateral papilledema. However, repeat magnetic resonance imaging of the brain with optic nerve windows failed to reveal enhancement of the optic nerves. Because the 2 lumbar punctures had shown normal opening pressure and because sequential ophthalmologic examination had revealed normal visual acuity and normal Goldmann visual fields, optic nerve edema was attributed to the high concentration of proteins in the cerebrospinal fluid. Before discharge, third magnetic resonance imaging of the head was performed and revealed 2 small white matter lesions, interpreted by the treating neurologic team as representing mild asymptomatic acute demyelinating encephalomyelitis.

In summary, case 4 presented with a typical clinical picture of postviral transverse myelitis and went on to develop superimposed Guillain-Barré syndrome and papilledema secondary to increased proteinorrachia. Imaging was remarkable for 2 asymptomatic white matter lesions suggestive of acute demyelinating encephalomyelitis. He recovered well with intravenous immunoglobulins.

Discussion

Central nervous system demyelination in conjunction with chronic inflammatory demyelinating polyradiculoneuropathy has been reported numerous times in the adult population.[1-19] Neuroimaging in affected patients has revealed cerebral demyelinating lesions in up to 23% of the patients studied.[8-12,17] Stojkovic et al[8] evaluated visual-evoked potentials in patients diagnosed as having chronic inflammatory demyelinating polyradiculoneuropathy and found abnormal potentials in 8 of 17 patients; 3 of these 8 patients had abnormal magnetic resonance imaging of the brain.

For the pediatric population, a review of the literature reveals 5 pediatric cases of concomitant chronic inflammatory demyelinating polyradiculoneuropathy and multiple sclerosis.[14,20,21] Pirko et al[20] described 3 patients with childhood-onset multiple sclerosis who developed chronic inflammatory demyelinating polyradiculoneuropathy several months after starting interferon therapy. Mariotti et al[21] and Watanabe et al[14] described patients who initially manifested central nervous system demyelinating symptoms, were treated with prednisone or intravenous immunoglobulins, and subsequently developed immunoglobulin-dependent chronic inflammatory demyelinating polyradiculoneuropathy in the absence of interferon therapy. Case 2 herein displays some similarities with these 3 patients (eg, chronic inflammatory demyelinating polyradiculoneuropathy associated with central nervous system demyelination), and the pathogenesis of their mixed disease may be similar.

There is scant literature on concomitant acute disseminated encephalomyelitis and Guillain-Barré syndrome in adult and pediatric populations. Okumura et al,[22] Amit et al,[23,24] and Itokazu et al[25] described patients with an infectious prodrome who developed central nervous system symptoms (coma, lethargy, and weakness) and were soon found to have absent or decreased reflexes and tetraparesis. In all of these cases, results of imaging, nerve conduction studies, and electromyography were consistent with combined demyelination of the central and peripheral nervous systems. Cerebrospinal fluid studies showed increased protein levels and negative oligoclonal banding. Therapy with corticosteroids or intravenous immunoglobulins induced recovery, with good outcome in all patients. These patients are similar to cases 1, 3, and 4 herein in terms of their presentation, response to therapy, and outcome, suggesting a common pathogenesis.

The occurrence of simultaneous demyelinating involvement of the peripheral and central nervous systems is poorly defined. It seems logical to assume an immunologic origin of these different diseases, and it has been suggested that an immune response to a common epitope component of both peripheral and central nervous system myelin could be responsible.[22,23] However, this does not explain the wide spectrum of combined involvement that is seen and its relative paucity among patients with demyelinating diseases. Further studies should better define these entities and may help explain the spectrum of severity and outcome seen with these diseases.

We believe that the disease in cases 1, 3, and 4 herein represents a distinct subgroup of demyelinating disease, predominantly seen in the pediatric population, manifesting as simultaneous demyelination of the peripheral and central nervous systems following an infectious insult. As previously noted among patients by Amit et al,[23] Okumura et al,[22] and Itokazu et al,[25] the patients herein responded well to corticosteroids (central nervous system demyelination) or intravenous immunoglobulins (peripheral demyelination) and had favorable outcomes, without recurrence.

Case 2 could be classified among pediatric and adult patients who present with evidence of a chronic central nervous system demyelinating disease with relapses (multiple sclerosis) and who subsequently develop chronic demyelination of the peripheral nervous system (chronic inflammatory demyelinating polyradiculoneuropathy) with or without interferon therapy. Outcome in this instance seems to be much more guarded, and requirement of long-term immune therapy is the rule for most patients. Our patient differs slightly from this pattern because she presented with mild central nervous system demyelinating disease and mild peripheral nervous system involvement, which evolved into chronic inflammatory demyelinating polyradiculoneuropathy without evidence of multiple sclerosis.

This classification is tentative as much is unknown about combined demyelinating diseases. For instance, one can hypothesize that systematic imaging in patients with Guillain-Barré syndrome or chronic inflammatory demyelinating polyradiculoneuropathy, as well as systematic neurophysiologic studies in patients with acute disseminated encephalomyelitis or transverse myelitis, would yield notable results. It may be that several demyelinating lesions go unnoticed, obscured by the clinical picture of the main disorder (eg, an altered level of consciousness in patients with acute disseminated encephalomyelitis can easily obscure a pattern of peripheral weakness or sensory symptoms).

Increasing knowledge of these pathologic conditions may trigger discovery of previously unrecognized patients and help better define the entities of Guillain-Barré syndrome, chronic inflammatory demyelinating polyradiculoneuropathy, acute disseminated encephalomyelitis, and multiple sclerosis in children. We believe that a multicenter prospective study of demyelinating disorders in children with strict criteria and systematic neurophysiologic and imaging studies would increase our understanding of the spectrum of the diseases and could lead to refinement of therapy (eg, corticosteroids vs immunoglobulins or plasma exchange in the acute setting, as well as the role of interferon therapy, in this pathologic condition).

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